

Study of the effect of correlated experimental uncertainty on Bayesian Inference of quark–gluon plasma properties

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Outline

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6. Next Steps

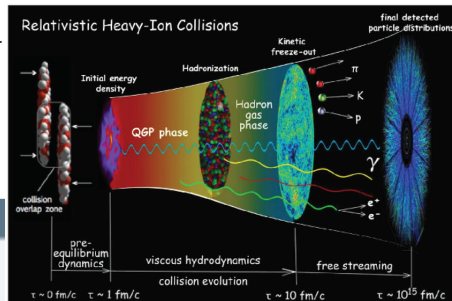
Motivation: Probing the Quark–Gluon Plasma

Relativistic heavy-ion collisions create extreme conditions where nuclear matter undergoes a transition to a deconfined state of quarks and gluons: the **Quark–Gluon Plasma (QGP)**.

Research

Constrain the properties of the Quark–Gluon Plasma by analyzing final-state hadronic observables and reconstructing the complete evolution of relativistic heavy-ion collisions.

- Initial-state conditions;
- Relativistic hydrodynamic;
- Particlization.



Schematic evolution of a relativistic heavy-ion collision.
Adapted from Purdue RCAC News (2023).

Baryon Enhancement

Observable

$$\frac{p}{\pi}(p_T) = \frac{dN_p/dp_T}{dN_\pi/dp_T}$$

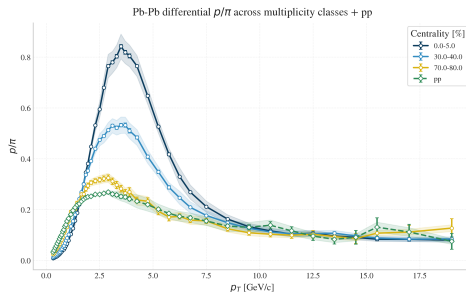
Enhancement region

$$2 \lesssim p_T \lesssim 5 \text{ GeV}/c$$

- Larger in central Pb–Pb.
- Driven by collective radial flow.
- Sensitive to QGP hadronization.

Data source: HEPData

10.17182/HEPData-ins1759506-v1.



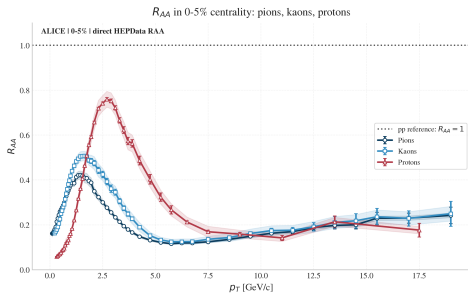
Quark coalescence

The high quark density in the QGP favors hadronization by quark recombination,

$$qqq \rightarrow p, \quad q\bar{q} \rightarrow \pi,$$

enhancing baryon production relative to mesons at intermediate p_T .

Nuclear Suppression



Jet quenching

Hard-scattered partons lose energy while traversing the QGP through multiple interactions with the medium. As a consequence, fewer high- p_T hadrons are reconstructed, leading to $R_{AA} < 1$.

Nuclear modification factor

$$R_{AA}(p_T) = \frac{1}{\langle N_{\text{coll}} \rangle} \frac{dN_{AA}/dp_T}{dN_{pp}/dp_T}$$

- $R_{AA} = 1$: binary collision scaling.
- $R_{AA} < 1$: suppression due to medium-induced energy loss.
- Intermediate p_T : protons are less suppressed than mesons.
- High p_T : all hadron species exhibit a similar suppression.

Bayesian Parameter Estimation

The properties of the QGP are not directly observable; they are inferred by comparing theoretical predictions with experimental measurements.

Bayes' theorem updates prior knowledge using experimental evidence:



$$P(\boldsymbol{\theta}|D) \propto P(D|\boldsymbol{\theta})P(\boldsymbol{\theta})$$

- $\boldsymbol{\theta}$: model parameters
- $P(\boldsymbol{\theta})$: prior
- $P(D|\boldsymbol{\theta})$: likelihood
- $P(\boldsymbol{\theta}|D)$: posterior

Experimental Data



Posterior samples are obtained using
MCMC.

J. E. Bernhard, J. S. Moreland and S. A. Bass, *Bayesian estimation of the specific shear and bulk viscosity of quark–gluon*

Phys. Rev. Lett. **115**, 1113–1117 (2015)

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Likelihood and Uncertainty Quantification

The agreement between model predictions and experimental data is quantified through a multivariate Gaussian likelihood:

$$\mathcal{L}(\theta) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp \left[-\frac{1}{2} (\mathbf{y}_m - \mathbf{y}_e)^T \Sigma^{-1} (\mathbf{y}_m - \mathbf{y}_e) \right]$$

where

$$\Sigma = \Sigma_{\text{exp}} + \Sigma_{\text{model}}.$$

Experimental uncertainty

$$\Sigma_{\text{exp}} = \Sigma_{\text{stat}}^{\text{exp}} + \Sigma_{\text{sys}}^{\text{exp}}$$

- Statistical uncertainty
- Correlated systematic uncertainty

Model uncertainty

$$\Sigma_{\text{model}} = \Sigma_{\text{emu}} + \Sigma_{\text{stat}}^{\text{model}} + \Sigma_{\text{sys}}^{\text{model}}$$

- Emulator prediction
- Simulation statistics
- Model discrepancy

Likelihood and Uncertainty Quantification

The posterior probability is obtained by maximizing the multivariate Gaussian likelihood.

$$\mathcal{L}(\theta) \propto \exp \left[-\frac{1}{2} (\mathbf{y}_m - \mathbf{y}_e)^T \Sigma^{-1} (\mathbf{y}_m - \mathbf{y}_e) \right]$$

$$\Sigma = \Sigma_{\text{exp}} + \Sigma_{\text{emu}} + \Sigma_{\text{model}}$$

Experimental covariance

$$\Sigma_{\text{exp}} = \Sigma_{\text{stat}} + \Sigma_{\text{sys}}$$

- Statistical uncertainty
- Correlated systematic uncertainty

Model covariance

$$\Sigma_{\text{model}} = \Sigma_{\text{pred}} + \Sigma_{\text{stat}}^{\text{model}} + \Sigma_{\text{sys}}^{\text{model}}$$

- Emulator prediction
- Simulation statistics
- Model discrepancy

Adapted from Bernhard *et al.*, Nature Physics 15 (2019).

Effect of the Experimental Covariance

Our study focuses on the experimental systematic covariance,

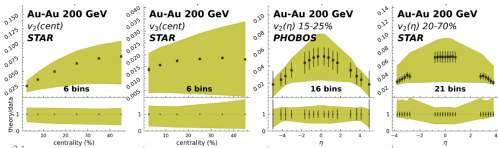
$$\Sigma_{\text{sys}}$$

which determines how systematic uncertainties are correlated among experimental data points.

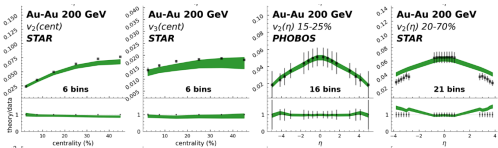
Different covariance models are investigated:

- Diagonal
- Compound Symmetry
- Radial Basis Function (RBF)

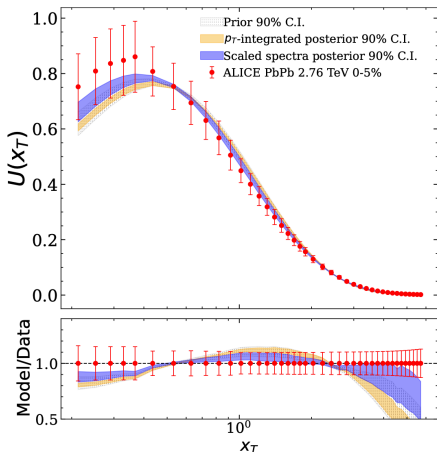
Changing Σ_{sys} modifies the



Bayesian calibration



Universal Scaled Spectra



Identified-particle spectra collapse onto a universal curve after rescaling.

Motivation

- Remove trivial dependence on particle yield.
- Remove the characteristic momentum scale.
- Compare only the **shape** of the spectra.

The transverse momentum is rescaled as

$$x_T = \frac{p_T}{\langle p_T \rangle},$$

and the corresponding scaled spectrum is

$$U(x_T) = \frac{\langle p_T \rangle}{N} \frac{dN}{dp_T} = \frac{1}{N} \frac{dN}{dx_T}.$$

After this normalization, spectra from different

Bayesian Calibration using Universal Scaled Spectra

Universal Scaled Spectra



Hydrodynamic Model

Calibration procedure

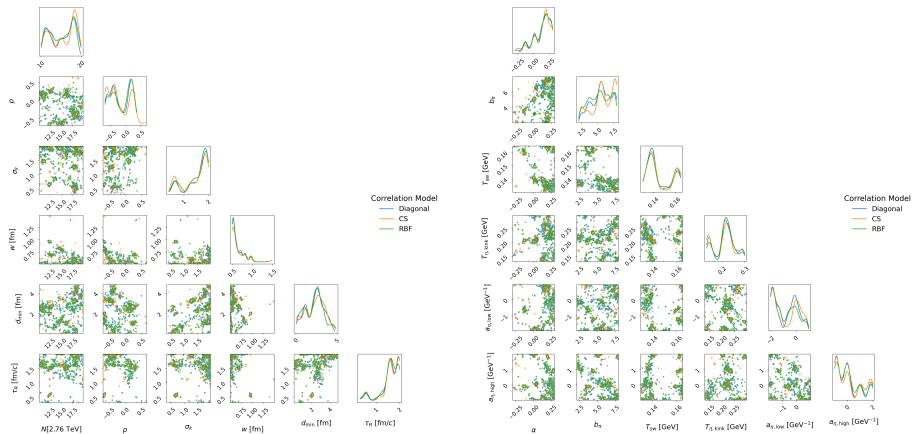
- Universal scaled spectra are used as experimental observables.
- Event-by-event hydrodynamic simulations are replaced by Gaussian Process emulators.
- Bayesian inference constrains the QGP model parameters.

This work

- Keep the original Bayesian calibration framework.
- Replace the diagonal experimental covariance matrix by correlated covariance models (CS and RBF).

Preliminary Results

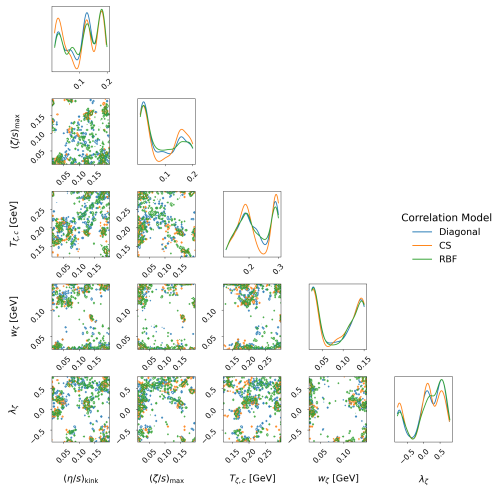
ALICE Pb–Pb $\sqrt{s_{NN}} = 2.76$ TeV, universal spectra calibration
HEPDData: ins1222333



Posterior distributions obtained using the Diagonal, Compound Symmetry (CS), and RBF experimental covariance models.

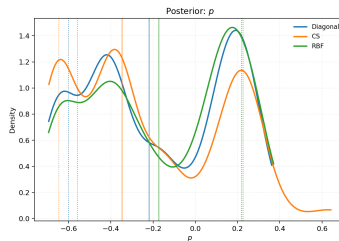
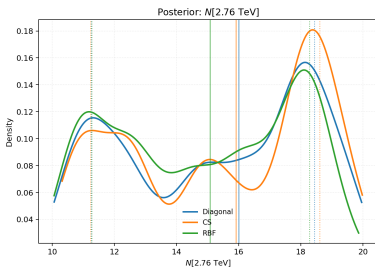
Preliminary Results

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Preliminary Results

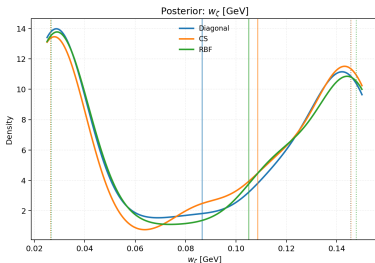


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parameters with
 $\Delta < 0.3\sigma$

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parameters with
 $\Delta > 0.3\sigma$



Largest posterior shifts

Parameter	Max shift
p	0.515σ
w_ζ	0.418σ
N	0.307σ

Largest normalized median shifts among the 17 inferred parameters.

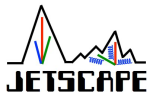
Changing the experimental covariance structure (Diagonal $\rightarrow$ CS $\rightarrow$ RBF) produces no statistically significant change in the inferred posterior for the Universal Scaled Spectra calibration, with posterior shifts remaining below 0.6σ across all inferred parameters.

Posterior comparison for ALICE Pb–Pb $\sqrt{s_{NN}} = 2.76 \text{ TeV}$, universal spectra calibration.

Next Steps

Planned extensions of this work

- Au–Au collisions at $\sqrt{s_{NN}} = 200$ GeV
- Multi-experiment calibration (STAR, PHENIX, PHOBOS and BRAHMS)
- Reproduce the JETSCAPE Bayesian analysis (Mankolli *et al.*, 2026) evaluating the impact of correlated experimental uncertainties arXiv:2601.17234



The JETSCAPE
Collaboration

Target reference

Thank you!



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